

# CIS241

## System-Level Programming and Utilities

### SSH

Erik Fredericks, [frederer@gvsu.edu](mailto:frederer@gvsu.edu)

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

# First, Linux

## What is it?

- Often referred to as an operating system (OS)
  - Competing with Mac/Windows
- *Technically*, it is a kernel

## Many OSs use the Linux kernel

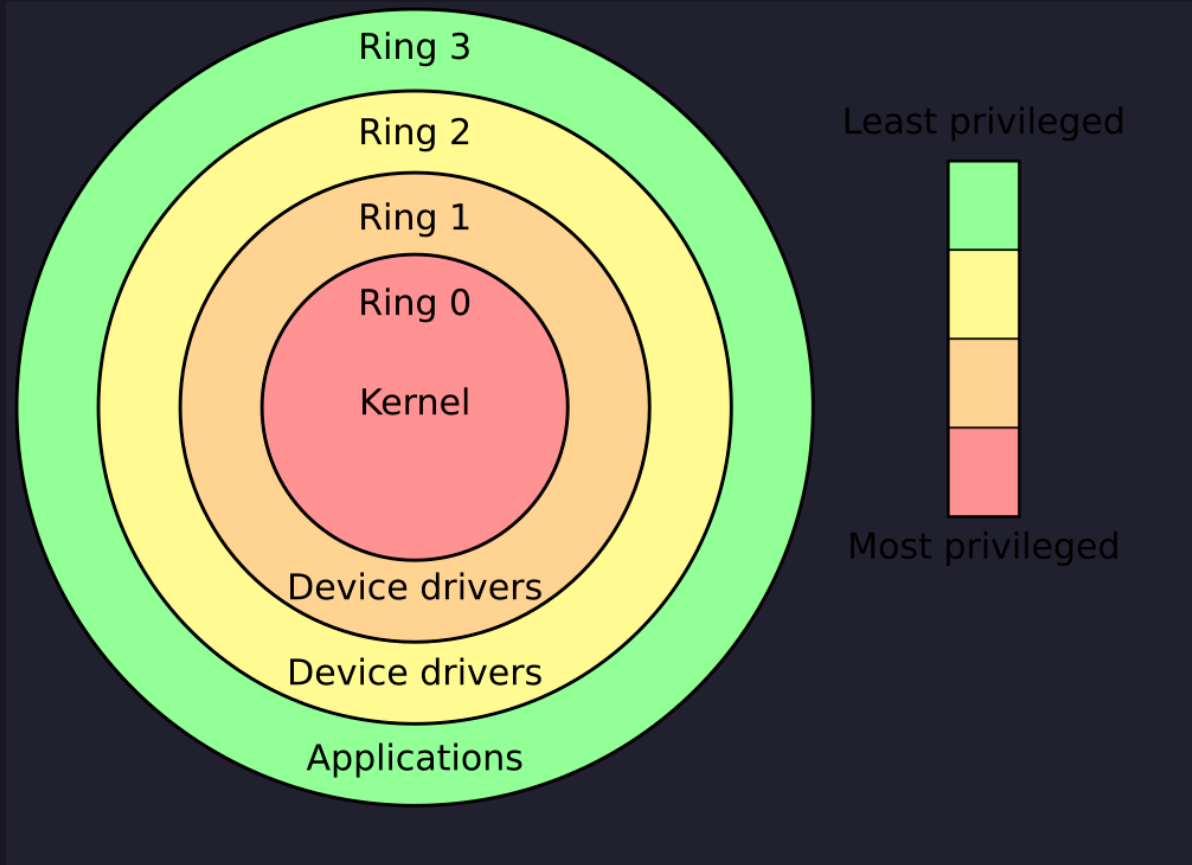
- Ubuntu (what is on EOS)
- Arch, Kali, Gentoo, Fedora, etc.

## Isn't Mac Linux too?

- Nope, Darwin! (Unix-based)



# Protection Rings



# A bit of history

**1970s: UNIX source code typically stripped from distribution**

- Exorbitant amounts of money to acquire source

**Linux first developed by Linus Torvalds to be a copy of UNIX**

- Released it to Internet for suggestions on how to improve it
- Kernel only updated by core developers and from suggestions
- Short history of Linux:
  - <https://cvw.cac.cornell.edu/linux/intro/history>



# Unix?

**Unix was an earlier operating system**

- Linux and other systems are heavily influenced by Unix

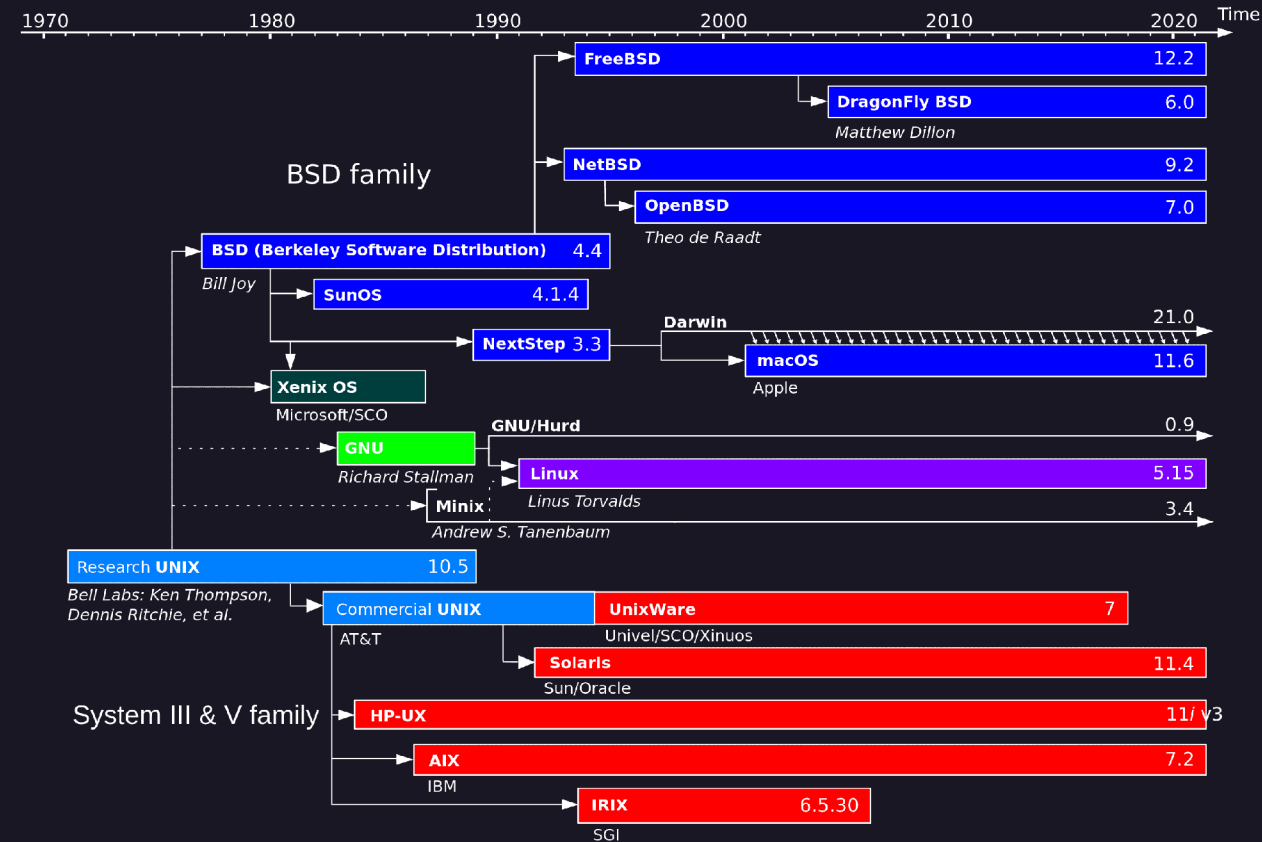
**We now use “Unix-like” to describe systems**

- They follow the same general ideas as Unix
- This means most things (commands, properties) carry over!

**"Unix-like" systems:**

- Linux
- macOS
- FreeBSD
- Solaris

# Unix Timeline



# Advantages of a Linux system

- Risk reduction
- Business needs
- Stability and security
- Flexibility
- Customization
- Support
- Cost reduction

# Advantages - Risk reduction

- Changes in the market or customer needs may cause companies to change software frequently
- Can be costly and time-consuming
- Support for closed source software may end
- Vendor may go out of business
- Software version may be retired
- OSS products offer the opportunity to maintain and change the source code



# Advantages - Business needs

- Common software available for Linux includes:
- Scientific and engineering software
- Software emulators
- Web servers, web browsers, and e-commerce suites
- Desktop productivity software
- Graphics manipulation software
- Database software
- Security software

# Advantages - Stability and security

- Customers using a closed source OS must rely on the OS vendor to fix any bugs
  - Waiting for a hot fix may take weeks or months
- The collaborative open source approach to testing and fixing bugs increases the stability of Linux
- Bugs and security loopholes in OSS programs can be identified and fixed quickly
  - Code is freely available and scrutinized by many developers

# Advantages - Flexibility

- Partial list of hardware platforms on which Linux can run:
  - Linux can be (and is) customized to work on mobile and embedded devices

|                   |             |
|-------------------|-------------|
| Intel x86/x64     | M68K        |
| Itanium           | PA-RISC     |
| Mainframe (S/390) | SPARC       |
| ARM               | Ultra-SPARC |
| Alpha             | PowerPC     |
| MIPS              |             |

# Advantage - Customization

## Ability to control the inner workings of the OS

- To use Linux as an Internet Web server, recompile the kernel to include only the support needed to be an Internet Web server
  - Results in a much smaller and faster kernel
- Can choose to install only software packages needed to perform required tasks
- Linux supports several programming languages, such as shell and PERL scripts to customize or automate tasks

# Advantage - Support

**Linux documentation can be easily found on the Internet**

- Frequently asked questions (FAQs)
- HOWTO documents

**HOWTO documents are maintained by their authors but are centrally collected by the Linux Documentation Project (LDP)**

**Linux newsgroups**

**Linux User Groups (LUGs): Open forum of Linux users who discuss and assist each other in using and modifying the Linux OS**

# Advantage - Cost reduction

## Linux is less expensive than other OSs

- There is no cost associated with acquiring the software
- A wealth of OSS can run on a variety of different hardware platforms running Linux

## The largest costs associated with Linux:

- Costs associated with hiring people to maintain the Linux system
- Potential support from vendors (i.e., Red Hat)

## Total cost of ownership (TCO): overall cost of using a particular OS

# Disadvantages of a Linux system

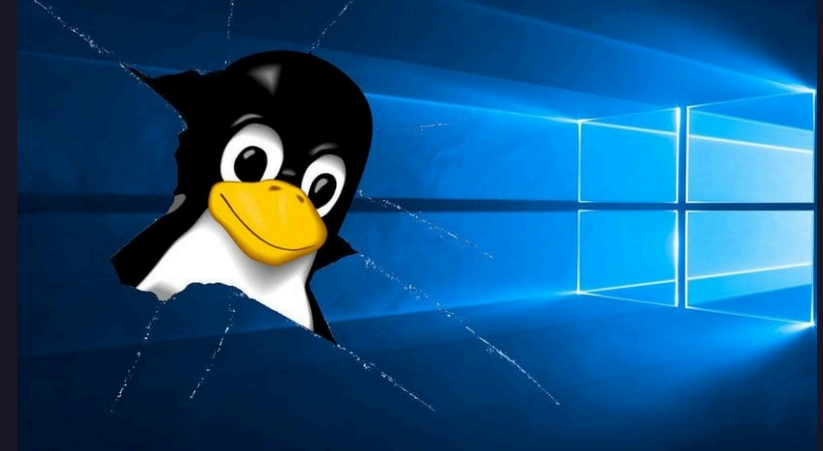
No standard edition of Linux

Support means that you'll be searching for very specific fixes

- Browsing SuperUser/ServerFault, forums, etc.

Not as easy to use as closed-source (Windows)

Program support



# Common uses of Linux

- Internet servers
- File/print server
- Application server
- Cloud servers
- Supercomputing
- Workstations (Office, personal, scientific, etc.)
- Mobile





# Windows

## Windows (today), is not Unix-like

- Can you do your homework on a Windows machine?
- Not by default

## However, we have WSL! (🐧\_🐧🌸)

- Windows Subsystem for Linux
- If you are on Windows, you must install WSL  
<https://learn.microsoft.com/en-us/windows/wsl/install>
  - Be very happy about this - I dreamed about just having the ability to SSH from the terminal when in industry

# Terminal

A screenshot of a terminal window with a dark background. On the left side, there is a large ASCII art logo made of green and yellow characters, resembling a stylized 'P' or a face. The main area of the terminal displays system information in a light blue/cyan font. At the top, it says 'erik@worktop' followed by a dashed line. Below that, it lists various system details: OS (Pop!\_OS 22.04 LTS x86\_64), Host (Latitude 7400), Kernel (Linux 6.12.10-76061203-generc), Uptime (2 days, 34 mins), Packages (4267 (dpkg), 7 (flatpak-us)), Shell (bash 5.1.16), Display (AU0243D): 1920x1080 @ 60 Hz], DE (GNOME 42.9), WM (Mutter (X11)), WM Theme (Pop-dark), Theme (Pop-dark [GTK2/3/4]), Icons (Pop [GTK2/3/4]), Font (Fira Sans (10pt, SemiLight) [G]), Cursor (Pop (24px)), Terminal (GNOME Terminal 3.44.0), Terminal Font (Fira Mono (12pt)), CPU (Intel(R) Core(TM) i7-8665U (8) z), GPU (Intel UHD Graphics 620 @ 1.15 G], Memory (15.70 GiB / 31.16 GiB (50%)), Swap (0 B / 20.00 GiB (0%)), Disk (/): 421.50 GiB / 928.94 GiB (44), Disk (/recovery): 2.84 GiB / 3.99 GiB, Local IP (wlo1): 192.168.1.46/24, Battery (DELL 5VC2M85): 100% [AC Con], and Locale (en\_US.UTF-8). At the bottom left, the prompt 'erik@worktop:~\$' is visible. At the bottom center, there is a small row of colored squares: red, green, yellow, blue, purple, cyan, and white.

# Shells (local)

See the little dollar sign?

Shell!

## What shell are we using?

```
echo $0
```

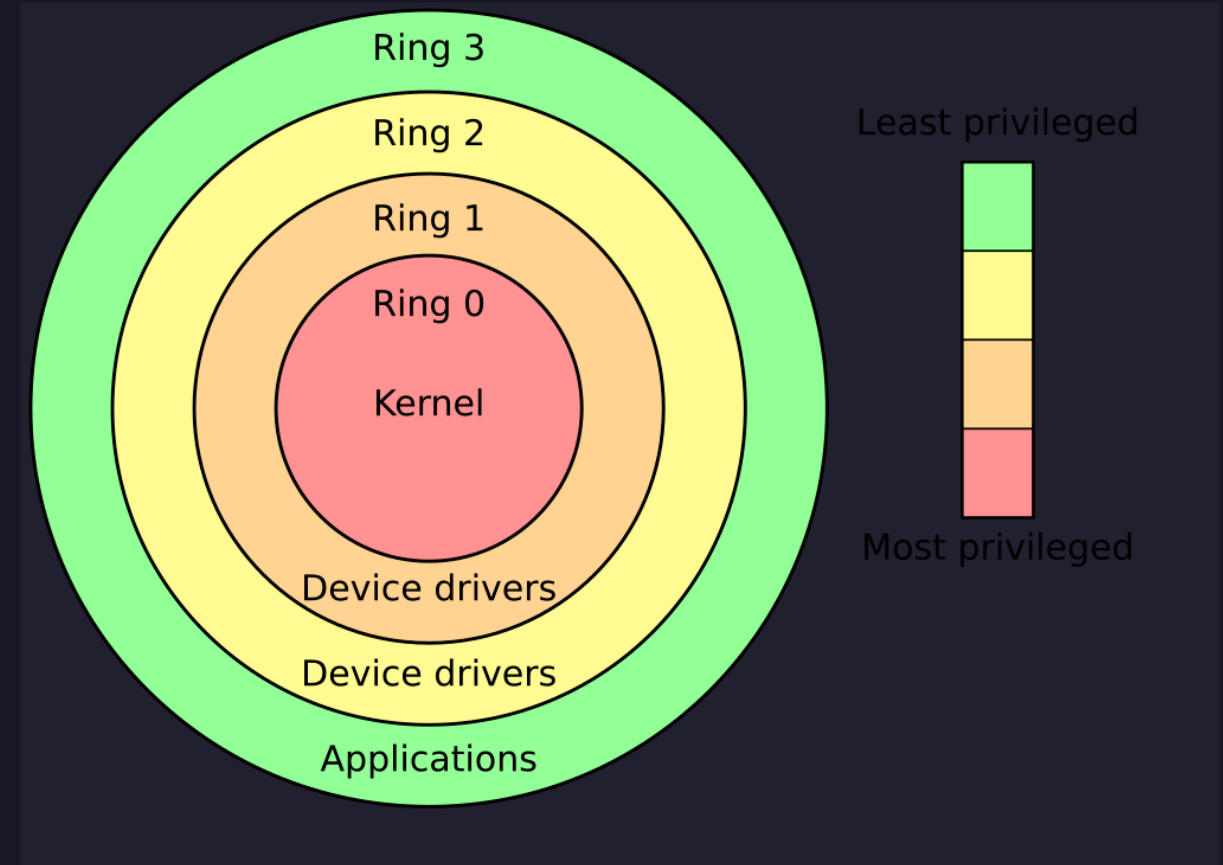
```
erik@worktop: ~  
  
/////////////////  
/////////////////  
767/////////////////  
767676*/////////////////  
/7676767/////////////////  
//*76767/////////////////  
///76767.///7676*/////////  
6//76767///767676/////////  
7676767///76767/////////  
6766,////////767/////////  
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erik@worktop  
-----  
OS: Pop!_OS 22.04 LTS x86_64  
Host: Latitude 7400  
Kernel: Linux 6.12.10-76061203-generc  
Uptime: 2 days, 34 mins  
Packages: 4267 (dpkg), 7 (flatpak-us)  
Shell: bash 5.1.16  
Display (AU0243D): 1920x1080 @ 60 Hz]  
DE: GNOME 42.9  
WM: Mutter (X11)  
WM Theme: Pop-dark  
Theme: Pop-dark [GTK2/3/4]  
Icons: Pop [GTK2/3/4]  
Font: Fira Sans (10pt, SemiLight) [G]  
Cursor: Pop (24px)  
Terminal: GNOME Terminal 3.44.0  
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CPU: Intel(R) Core(TM) i7-8665U (8) z  
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Memory: 15.70 GiB / 31.16 GiB (50%)  
Swap: 0 B / 20.00 GiB (0%)  
Disk (/): 421.50 GiB / 928.94 GiB (44  
Disk (/recovery): 2.84 GiB / 3.99 GiB  
Local IP (wlo1): 192.168.1.46/24  
Battery (DELL 5VC2M85): 100% [AC Con]  
Locale: en_US.UTF-8  
  
$ █
```

# Shells (local)

Where is the shell here?

- Ring 3, typically!

How do we run in other rings?



You can use `eos01.cis.gvsu.edu` - `eos30.cis.gvsu.edu`!

## Shells (remote)

Note - if you're off-campus you'll need a VPN!

Or, connecting to another machine

- Or or, **doing your homework** ټ\_ټ

`ssh` --> Secure Shell

Usage: `ssh username@address`

For example, connecting to EOS:

`ssh username@eos1.cis.gvsu.edu`

Lab fun

DO NOT

SHUT DOWN

THESE MACHINES



# Last - the web interface

<https://computing.gvsu.edu>

This will provide a web-based point of access for you!

- **Allows for desktop or terminal**
- **Note that the user experience is not great!**
  - Can't copy/paste files from your computer to the desktop
  - Random connection breaks common
- **However - don't need a VPN!**

# Ok now really the last

To exit an SSH session (or to kill your terminal)

```
exit
```